MTH251 & MTH252

Mathematical Analysis I & II

Course Textbook

Department of Mathematical Sciences

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Mathematical Analysis I & II

Course Textbook

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Source Code & Contributions
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Chapter 1

The Number Systems

1.1 The Natural, Integer, and Rational Number Systems

We denote the set $\{1, 2, 3, \ldots\}$ of natural numbers by $\mathbb{N}$. The set $\mathbb{N}$ is defined by the following axioms:

- $1 \in \mathbb{N}$.
- If $n \in \mathbb{N}$, then $s(n) \in \mathbb{N}$ where $s(n) = n + 1$ is called the **successor** of n.
- 1 is not the successor of any element of $\mathbb{N}$.
- If $m, n \in \mathbb{N}$ and $s(m) = s(n)$, then $m = n$.
- Any subset of $\mathbb{N}$ which contains 1, and contains $s(n)$ whenever it contains n, must be equal to $\mathbb{N}$.

The axioms above are called the **Peano axioms** (or *Peano Postulates*).

Remark. Let $P(n)$ be a statement about $n \in \mathbb{N}$ and $S := \{n \in \mathbb{N} \mid P(n) \text{ is true}\}$. Suppose $P(n)$ satisfies the following conditions:

- $P(1)$ is true.
- $P(n + 1)$ is true whenever $P(n)$ is true.

This implies that

- $1 \in S$.
- $n + 1 \in S$ whenever $n \in S$.

Thus $S = \mathbb{N}$, i.e., $P(n)$ is true for all $n \in \mathbb{N}$. This method of proof is called the **principle of mathematical induction**.

Theorem 1.1.1. *Every nonempty subset of $\mathbb{N}$ has a least element, i.e., if $E \subseteq \mathbb{N}$ and $E \neq \emptyset$, then there exists an element $m \in E$ such that*

$$m \leq n \quad \text{for all } n \in E.$$

This property is called the ***well-ordering property*** *of $\mathbb{N}$.*

Proof. Let S be a nonempty subset of $\mathbb{N}$ which does not have a least element. We define a set $T := \mathbb{N} \setminus S$, and $L := \{n \in \mathbb{N} \mid \{1, 2, \ldots, n\} \subseteq T\}$. We will show that $L = \mathbb{N}$, which implies that $S = \emptyset$, a contradiction, by mathematical induction.

- **Base case:** $1 \in L$ since $\{1\} \subseteq T$, otherwise $1 \in S$ and 1 is the least element of S.
- **Inductive step:** Suppose $n \in L$, i.e., $\{1, 2, \ldots, n\} \subseteq T$. Then $n + 1 \notin S$, otherwise $n + 1$ is the least element of S. Therefore, $n + 1 \in T$, and $\{1, 2, \ldots, n + 1\} \subseteq T$, i.e., $n + 1 \in L$.

By the principle of mathematical induction, $L = \mathbb{N}$. $\square$

Remark. Now, we can define the set $\mathbb{Z}$ of integers and the set $\mathbb{Q}$ of rational numbers as follows:

- $\mathbb{Z} := \{m - n \mid m, n \in \mathbb{N}\}$.
- $\mathbb{Q} := \left\{ \frac{m}{n} \;\middle|\; m \in \mathbb{Z},\ n \in \mathbb{N} \right\}$.

Also, we denote the set of positive numbers or negative numbers as follows:

- $\mathbb{Z}^+ := \{n \mid n \in \mathbb{N}\}$, $\mathbb{Z}^- := \{-n \mid n \in \mathbb{N}\}$.
- $\mathbb{Q}^+ := \left\{ \frac{m}{n} \;\middle|\; m \in \mathbb{N},\ n \in \mathbb{N} \right\}$, $\mathbb{Q}^- := \left\{ \frac{m}{n} \;\middle|\; m \in \mathbb{Z}^-,\ n \in \mathbb{N} \right\}$.

1.2 Ordered Sets

An **ordered set** is a set S together with a relation $<$ such that for any $x, y, z \in S$,

(A) $x < y$ and $y < z$ implies $x < z$.

(B) Only one of the relations $x < y$, $x = y$, or $x > y$ holds.

$x > y$ means $y < x$, $x \leq y$ means ($x < y$ or $x = y$) and $x \geq y$ means $y \leq x$.

Remark. In an ordered set S, the relation $<$ is called a **order** (or a **total order**) on S and the set S is called **totally ordered**. If a relation $<$ satisfies **(A)** only, then the relation $<$ is called a **partial order** on S and the set S is called **partially ordered**.

Example 1.2.1. The set $\mathbb{N}$ and $\mathbb{Z}$ are ordered sets, where $m > n \iff m - n \in \mathbb{N}$. The set $\mathbb{Q}$ is also an ordered set, where $m > n \iff m - n \in \mathbb{Q}^+$.

Definition 1.2.2. Let S be an ordered set and $E \subseteq S$.

- E is **bounded above** if there exists $M \in S$ such that $x \leq M$ for all $x \in E$. Such an M is called an **upper bound** of E.
- E is **bounded below** if there exists $m \in S$ such that $m \leq x$ for all $x \in E$. Such an m is called a **lower bound** of E.

Theorem 1.2.3. *Any nonempty subset of $\mathbb{Z}$ which is bounded below has a least element, and any nonempty subset of $\mathbb{Z}$ which is bounded above has a greatest element.*

Proof. Let $E \subseteq \mathbb{Z}$ be nonempty and bounded below. Then there exists $m \in \mathbb{Z}$ such that $m \leq x$ for all $x \in E$. Let $F := \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}$. By Theorem 1.1.1, F has a least element n.

$$n \leq x - m + 1 \text{ for all } x \in E \implies n + m - 1 \leq x \text{ for all } x \in E.$$

Since $n \in F$, there exists $x_0 \in E$ such that $n = x_0 - m + 1$, so $x_0 = n + m - 1$ is the least element of E.

Let $S \subseteq \mathbb{Z}$ be nonempty and bounded above. Then there exists $M \in \mathbb{Z}$ such that $x \leq M$ for all $x \in S$. Let $T := \{-x + M + 1 \mid x \in S\} \subseteq \mathbb{N}$. By Theorem 1.1.1, T has a least element n.

$$n \leq -x + M + 1 \text{ for all } x \in S \implies M - n + 1 \geq x \text{ for all } x \in S.$$

Since $n \in T$, there exists $x_0 \in S$ such that $n = -x_0 + M + 1$, so $x_0 = M - n + 1$ is the greatest element of S. $\square$

Definition 1.2.4. Let S be an ordered set and $E \subseteq S$. Suppose E is bounded above. Suppose there exists an $\alpha \in S$ such that

- α is an upper bound of E.
- If $\beta < \alpha$, then β is not an upper bound of E.

Then α is called the *least upper bound* or **supremum** of E, denoted by

$$\alpha = \sup E.$$

Similarly, suppose E is bounded below. Suppose there exists a $\alpha \in S$ such that

- α is a lower bound of E.
- If $\beta > \alpha$, then β is not a lower bound of E.

Then α is called the *greatest lower bound* or **infimum** of E, denoted by

$$\alpha = \inf E.$$

Example 1.2.5. Let $S := \{p \in \mathbb{Q}^+ \mid p^2 < 2\}$. For any $p \in S$, let $q = p - \dfrac{p^2-2}{p+2} = \dfrac{2p+2}{p+2} > p$. Then $q^2 - 2 = \dfrac{2(p^2-2)}{(p+2)^2} < 0$, so $q \in S$. This implies that any $p \in S$ is not an upper bound of S.

For any $p \in \mathbb{Q}^+$ with $p^2 > 2$, p is an upper bound of S. Let $q = p - \dfrac{p^2-2}{p+2} = \dfrac{2p+2}{p+2} < p$. Then $q \in \mathbb{Q}^+$ and $q^2 - 2 = \dfrac{2(p^2-2)}{(p+2)^2} > 0$, so q is an upper bound of S. This implies that such p is not the least upper bound of S.

Since there exists no $p \in \mathbb{Q}$ such that $p^2 = 2$, the set S has no least upper bound in $\mathbb{Q}^+$.

Definition 1.2.6. An ordered set S is said to have the **least upper bound property** if every nonempty subset $E \subseteq S$ that is bounded above has a least upper bound in S.

Theorem 1.2.7. *If S is an ordered set with the least upper bound property, then any nonempty subset $E \subseteq S$ that is bounded below has a greatest lower bound in S.*

Proof. Let E be a nonempty subset of S that is bounded below. Let L be the set of all lower bounds of E. Then L is nonempty and bounded above by any element of E. Let $\alpha = \sup L$. We claim that $\alpha = \inf E$.

Suppose $\alpha \notin L$. Then there exists $\beta \in E$ such that $\beta < \alpha$. Since $\alpha = \sup L$, there exists $\gamma \in L$ such that $\beta < \gamma \leq \alpha$. But γ is a lower bound of E, so $\gamma \leq \beta$, a contradiction. Therefore, $\alpha \in L$.

Since $\alpha = \sup L$, for any $\delta > \alpha$, $\delta \notin L$. Therefore, $\alpha = \inf E$. $\square$

1.3 Fields and the Real Field

A set F is called a **field** if it is equipped with two operations, *addition* and *multiplication*, and satisfies the following axioms:

(A) Axioms for addition

- $x + y \in F$ for all $x, y \in F$.
- $x + y = y + x$ for all $x, y \in F$.
- $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- There exists an element $0 \in F$ such that $x + 0 = x$ for all $x \in F$.
- For each $x \in F$, there exists an element $-x \in F$ such that $x + (-x) = 0$.

(B) Axioms for multiplication

- $x \cdot y \in F$ for all $x, y \in F$.
- $x \cdot y = y \cdot x$ for all $x, y \in F$.
- $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in F$.
- There exists an element $1 \in F$ with $1 \neq 0$ such that $x \cdot 1 = x$ for all $x \in F$.
- For each $x \in F$ with $x \neq 0$, there exists an element $x^{-1} \in F$ such that $x \cdot x^{-1} = 1$.

(C) The distributive law

- $x \cdot (y + z) = x \cdot y + x \cdot z$ for all $x, y, z \in F$.

Remark. A set F is called an **ordered field** if it is both an ordered set and a field, and satisfies the following axioms:

- If $x, y, z \in F$ and $x < y$, then $x + z < y + z$.
- If $x, y \in F$ and $x > 0$ and $y > 0$, then $x \cdot y > 0$.

Remark. There exists an ordered field $\mathbb{R}$ which has the least upper bound property. Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield, i.e., $\mathbb{Q} \subseteq \mathbb{R}$ and the operations of addition and multiplication on $\mathbb{R}$ coincide with those on $\mathbb{Q}$.

We can construct $\mathbb{R}$ from $\mathbb{Q}$. But for now, we will assume the existence of $\mathbb{R}$ and its properties.

Theorem 1.3.1. *If $x, y \in \mathbb{R}$ and $x > 0$, then there exists some $n \in \mathbb{N}$ such that $nx > y$. This theorem is called the **Archimedean property** of $\mathbb{R}$.*

Proof. Suppose $x, y \in \mathbb{R}$ and $x > 0$. Let $E := \{nx \mid nx \leq y,\ n \in \mathbb{N}\}$. If $E = \emptyset$, which implies that $x > y$, $n = 1$ satisfies $nx > y$. Otherwise, E is nonempty and bounded above by y.

By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}$. Then $\alpha - x$ is not an upper bound of E. Therefore, there exists some $m \in \mathbb{N}$ such that $mx > \alpha - x \implies (m+1)x > \alpha$. Then $(m+1)x \notin E$, which implies that $(m+1)x > y$. Therefore, we can take $n = m + 1$. □

Theorem 1.3.2. *If $x, y \in \mathbb{R}$ and $x < y$, then there exists some $p \in \mathbb{Q}$ such that $x < p < y$. This theorem states that $\mathbb{Q}$ is **dense** in $\mathbb{R}$.*

Proof. Let $\epsilon = y - x > 0$. By Theorem 1.3.1, there exists some $n \in \mathbb{N}$ such that $\dfrac{1}{n} < \epsilon$. Let $S := \left\{k \in \mathbb{Z} \;\middle|\; \dfrac{k}{n} < x\right\}$. By Theorem 1.2.3, there exists $m = \max S \in S$. Then we have

$$x < \frac{m+1}{n} < x + \frac{1}{n} < x + (y - x) = y.$$

Therefore, we can take $p = \dfrac{m+1}{n} \in \mathbb{Q}$. □

Theorem 1.3.3. *For each $x \in \mathbb{R}^+$ and $n \in \mathbb{N}$, there exists a unique $y \in \mathbb{R}^+$ such that $y^n = x$. We denote y by $\sqrt[n]{x}$ or $x^{\frac{1}{n}}$.*

Proof. Let $E := \{t \in \mathbb{R}^+ \mid t^n < x\}$. Then E is nonempty since $2^{-1} \in E$ if $x \geq 1$, and $x^2 \in E$ otherwise. Moreover, E is bounded above by $\max\{x, 1\}$. By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}^+$. We will show that $\alpha^n = x$.

For any $a, b \in \mathbb{R}^+$ with $a < b$, we have

$$b^n - a^n = (b - a)\sum_{k=0}^{n-1} b^{n-1-k}a^k < (b - a)nb^{n-1}.$$

Assume that $\alpha^n < x$. By Theorem 1.3.2, we can choose $h \in \mathbb{R}^+$ such that

$$h < \min\left\{1,\ \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}}\right\}.$$

Put $a = \alpha$ and $b = \alpha + h$. Then we have

$$\begin{aligned} &(\alpha + h)^n - \alpha^n < hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies &(\alpha + h)^n < \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x. \end{aligned}$$

This shows that $\alpha + h \in E$, which contradicts the fact that $\alpha = \sup E$.

Assume that $\alpha^n > x$. By Theorem 1.3.2, we can choose $h \in \mathbb{R}^+$ such that

$$h < \min\left\{1,\ \alpha,\ \frac{\alpha^n - x}{n(\alpha+1)^{n-1}}\right\}.$$

Put $a = \alpha - h$ and $b = \alpha$. Then we have

$$\begin{aligned} &\alpha^n - (\alpha - h)^n < hn\alpha^{n-1} < hn(\alpha+1)^{n-1} \\ \implies &(\alpha - h)^n > \alpha^n - hn(\alpha+1)^{n-1} > \alpha^n - (\alpha^n - x) = x. \end{aligned}$$

This shows that $\alpha - h$ is an upper bound of E, which contradicts the fact that $\alpha = \sup E$.

Therefore, we have $\alpha^n = x$, which guarantees the *existence* of y. For any $\alpha, \beta \in \mathbb{R}^+$, $\alpha \neq \beta \implies \alpha^n \neq \beta^n$. Thus the *uniqueness* of y is guaranteed. $\square$

Remark. Theorem 1.2.3 supposes that E is bounded in $\mathbb{Z}$. By Theorem 1.3.1, we can always find an integer k such that $k < x$ or $x < k$ for any $x \in \mathbb{R}$. Therefore, Theorem 1.2.3 also holds when E is bounded in $\mathbb{R}$, which is followed by the fact that E is also bounded in $\mathbb{Z}$.

Definition 1.3.4. For any $x \in \mathbb{R}$, the **floor function** $\lfloor x \rfloor$ and **ceiling function** $\lceil x \rceil$ are defined as follows:

$$\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}, \quad \lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$$

whose existences are guaranteed by Theorem 1.2.3.

Theorem 1.3.5. *Let $x \in \mathbb{R}$ and $b \in \mathbb{N}$ be given with $b \geq 2$. For any $n \in \mathbb{N}$, there exists a unique sequence of integers $\{a_1, a_2, \ldots, a_n\}$ such that*

$$z_0 \leq x - \sum_{k=1}^{n} \frac{a_k}{b^k} < z_0 + \frac{1}{b^n}$$

*where $z_0 = \lfloor x \rfloor$ and $a_k \in S = \{0, 1, 2, \ldots, b-1\}$ for all $k = 1, 2, \ldots, n$. The sequence $\{z_0, a_1, a_2, \ldots, a_n\}$ is called the **base** b **expansion** of x. Especially, if $b = 10$, then the sequence is called the **decimal expansion** of x.*

Proof. Let $T_n \coloneqq \left\{ k \in S \;\middle|\; z_n + \dfrac{k}{b} \le b^n x \right\}$ for $n \in \mathbb{N} \cup \{0\}$, where $z_n = \lfloor b^n x \rfloor$. Since $0 \in T_n$ and T_n is bounded above, by Theorem 1.2.3, there exists $a_n = \max T_{n-1} \in T_{n-1}$ for each $n \in \mathbb{N}$.

First, we will show the *existence* by mathematical induction on n. Let P_n : The sequence of integers $\{a_1, a_2, \ldots, a_n\}$ satisfies

$$z_0 \le x - \sum_{k=1}^{n} \frac{a_k}{b^k} < z_0 + \frac{1}{b^n}.$$

- **Base case:** Since $a_1 = \max T_0 \in T_0$, we have

$$z_0 + \frac{a_1}{b} \le x < z_0 + \frac{a_1 + 1}{b} \implies z_0 \le x - \frac{a_1}{b} < z_0 + \frac{1}{b}.$$

  Thus P_1 is true.

- **Inductive step:** Assume that P_n is true for some $n \in \mathbb{N}$. Then we have

$$\begin{aligned}
&z_0 \le x - \sum_{k=1}^{n} \frac{a_k}{b^k} < z_0 + \frac{1}{b^n} \\
\implies\ &z_0 b^n \le b^n x - \sum_{k=1}^{n} a_k b^{n-k} < z_0 b^n + 1 \\
\implies\ &z_0 b^n + \sum_{k=1}^{n} a_k b^{n-k} \le b^n x < z_0 b^n + \sum_{k=1}^{n} a_k b^{n-k} + 1.
\end{aligned}$$

  By Definition 1.3.4, we have

$$z_n = z_0 b^n + \sum_{k=1}^{n} a_k b^{n-k}.$$

  Since $a_{n+1} = \max T_n \in T_n$, we have

$$\begin{aligned}
&z_n + \frac{a_{n+1}}{b} \le b^n x < z_n + \frac{a_{n+1} + 1}{b} \\
\implies\ &z_0 b^n + \sum_{k=1}^{n} a_k b^{n-k} + \frac{a_{n+1}}{b} \le b^n x < z_0 b^n + \sum_{k=1}^{n} a_k b^{n-k} + \frac{a_{n+1} + 1}{b} \\
\implies\ &z_0 b^n + \sum_{k=1}^{n+1} a_k b^{n-k} \le b^n x < z_0 b^n + \sum_{k=1}^{n+1} a_k b^{n-k} + \frac{1}{b} \\
\implies\ &z_0 + \sum_{k=1}^{n+1} \frac{a_k}{b^k} \le x < z_0 + \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \\
\implies\ &z_0 \le x - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < z_0 + \frac{1}{b^{n+1}}.
\end{aligned}$$

  Thus P_{n+1} is true whenever P_n is true.

By the principle of mathematical induction, P_n is true for all $n \in \mathbb{N}$. Thus the *existence* of the sequence $\{a_1, a_2, \ldots, a_n\}$ for each $n \in \mathbb{N}$ is guaranteed.

Next, we will show the *uniqueness* of the sequence. Suppose that there exist two sequences of integers $\{a_1, a_2, \ldots, a_n\}$ and $\{c_1, c_2, \ldots, c_n\}$ such that

- $z_0 \leq x - \sum_{k=1}^{n} \frac{a_k}{b^k} < z_0 + \frac{1}{b^n}$.
- $z_0 \leq x - \sum_{k=1}^{n} \frac{c_k}{b^k} < z_0 + \frac{1}{b^n}$.

for some $n \in \mathbb{N}$. Then we have both

- $\sum_{k=1}^{n} \frac{a_k}{b^k} \leq x - z_0 < \sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n}$.
- $\sum_{k=1}^{n} \frac{c_k}{b^k} \leq x - z_0 < \sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n}$.

By combining the two inequalities, we have

$$\sum_{k=1}^{n} \left| \frac{a_k - c_k}{b^k} \right| < \frac{1}{b^n} \implies \left| \frac{a_k - c_k}{b^k} \right| < \frac{1}{b^n} \text{ for all } k = 1, 2, \ldots, n$$
$$\implies |a_k - c_k| < b^{k-n} \leq 1 \text{ for all } k = 1, 2, \ldots, n.$$

Since $a_k, c_k \in S$ and so is $|a_k - c_k|$, we have $a_k = c_k$ for all $k = 1, 2, \ldots, n$. Thus the *uniqueness* of the sequence $\{a_1, a_2, \ldots, a_n\}$ for each $n \in \mathbb{N}$ is guaranteed. $\square$

1.4 The Extended Real Number System

The Extended real number system $\overline{\mathbb{R}}$ is defined as

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

We define the following operations on $\overline{\mathbb{R}}$:

- $x + (+\infty) = +\infty$ for all $x \in \mathbb{R} \cup \{\infty\}$.
- $x + (-\infty) = -\infty$ for all $x \in \mathbb{R} \cup \{-\infty\}$.
- $x \cdot (\pm\infty) = \pm\infty$ for all $x \in \mathbb{R}^+ \cup \{\infty\}$.
- $x \cdot (\pm\infty) = \mp\infty$ for all $x \in \mathbb{R}^- \cup \{-\infty\}$.
- $\frac{x}{\pm\infty} = 0$ for all $x \in \mathbb{R}$.

$\overline{\mathbb{R}}$ is not a field, but it is an ordered set with the following order relation:

$$-\infty < x < +\infty \quad \text{for all } x \in \mathbb{R}.$$

Theorem 1.4.1. *Every subset of $\overline{\mathbb{R}}$ has a least upper bound and a greatest lower bound in $\overline{\mathbb{R}}$.*

Proof. Let $E \subseteq \overline{\mathbb{R}}$. If $E = \emptyset$, then $\sup E = -\infty$ and $\inf E = +\infty$. If $E \neq \emptyset$, then we have the following cases:

- If E is bounded above in $\mathbb{R}$, then $\sup E$ exists in $\mathbb{R}$ by the least upper bound property of $\mathbb{R}$.
- If E is not bounded above in $\mathbb{R}$, then $\sup E = +\infty$.
- If E is bounded below in $\mathbb{R}$, then $\inf E$ exists in $\mathbb{R}$ by the greatest lower bound property of $\mathbb{R}$.
- If E is not bounded below in $\mathbb{R}$, then $\inf E = -\infty$.

□

1.5 The Complex Field

The complex field $\mathbb{C}$ is defined as

$$\mathbb{C} \coloneqq \{(x, y) \mid x, y \in \mathbb{R}\}.$$

We define the following operations on $\mathbb{C}$:

- $(x, y) + (u, v) = (x + u, y + v)$ for all $(x, y), (u, v) \in \mathbb{C}$.
- $(x, y) \cdot (u, v) = (xu - yv, xv + yu)$ for all $(x, y), (u, v) \in \mathbb{C}$.

$\mathbb{C}$ is a field with the following properties:

- The additive identity is $(0, 0)$ and the multiplicative identity is $(1, 0)$.
- The additive inverse of (x, y) is $(-x, -y)$ and the multiplicative inverse of (x, y) is
$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2}\right)$$
for all $(x, y) \in \mathbb{C}$ with $(x, y) \neq (0, 0)$.

Remark. The function

$$f : \{(x, 0) \mid x \in \mathbb{R}\} \subseteq \mathbb{C} \to \mathbb{R}$$

defined by $f((x, 0)) = x$ is an *isomorphism*. Then we can identify $\mathbb{R}$ with the subfield $\{(x, 0) \mid x \in \mathbb{R}\}$ of $\mathbb{C}$. Now we will denote $(x, 0) \in \mathbb{C}$ by $x \in \mathbb{R}$.

Remark. We define $i = (0, 1) \in \mathbb{C}$. Then we have $i^2 = -1$, and any complex number $(x, y) \in \mathbb{C}$ can be written as

$$(x, y) = x + yi.$$

Definition 1.5.1. Let $z = x + yi \in \mathbb{C}$ for some $x, y \in \mathbb{R}$. The **real part** of z is defined as

$$\Re(z) = x \text{ or } \mathrm{Re}(z) = x,$$

the **imaginary part** of z is defined as

$$\Im(z) = y \text{ or } \mathrm{Im}(z) = y,$$

and the **complex conjugate** of z is defined as

$$\overline{z} = x - yi.$$

Theorem 1.5.2. *Let $z, w \in \mathbb{C}$. Then we have the following properties:*

- $\overline{(\overline{z})} = z$.
- $z = \overline{z} \iff z \in \mathbb{R}$.
- $\overline{z + w} = \overline{z} + \overline{w}$.
- $\overline{zw} = \overline{z} \cdot \overline{w}$.
- $z + \overline{z} = 2\mathrm{Re}(z)$ *and* $z - \overline{z} = 2i\mathrm{Im}(z)$.
- $z\overline{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}$.

Proof. The proof follows directly from the definitions and properties of the operations on $\mathbb{C}$. □

Definition 1.5.3. Let $z \in \mathbb{C}$. the **modulus** of z is defined as

$$|z| = \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}.$$

The *existence* and *uniqueness* of $|z|$ follows from the last part of Theorem 1.5.2 and Theorem 1.3.3.

Theorem 1.5.4. *Let $a_1, a_2, \dots, a_n \in \mathbb{C}$ and $b_1, b_2, \dots, b_n \in \mathbb{C}$. Then we have*

$$\left|\sum_{k=1}^{n} a_k \overline{b_k}\right|^2 \leq \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2.$$

This theorem is called the ***Cauchy-Schwarz inequality****.*

Proof. Let $i, j \in \mathbb{N}$ with $1 \leq i < j \leq n$. Then we have

$$\begin{aligned} &|a_i b_j - a_j b_i|^2 = (a_i b_j - a_j b_i)\overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}). \end{aligned}$$

Consider the following sums:

$$\begin{aligned} &\sum_{i<j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) = \sum_{i<j} (|a_i|^2 |b_j|^2) + \sum_{j<i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) = \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \sum_{k=1}^{n} |a_k|^2 |b_k|^2. \end{aligned}$$

$$\begin{aligned} &\sum_{i<j} (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}) = \sum_{i \neq j} (a_i \overline{b_i} \cdot \overline{a_j} b_j) \\ &= \sum_{i=1}^{n} a_i \overline{b_i} \cdot \sum_{j=1}^{n} \overline{a_j} b_j - \sum_{k=1}^{n} a_k \overline{b_k} \cdot \overline{a_k} b_k = \left|\sum_{k=1}^{n} a_k \overline{b_k}\right|^2 - \sum_{k=1}^{n} |a_k|^2 |b_k|^2. \end{aligned}$$

Therefore, we have

$$\sum_{i<j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^{n} |a_k|^2 \cdot \sum_{k=1}^{n} |b_k|^2 - \left|\sum_{k=1}^{n} a_k \overline{b_k}\right|^2 \geq 0.$$

□

Corollary 1.5.5. *Let $a_1, a_2, \dots, a_n \in \mathbb{R}$ and $b_1, b_2, \dots, b_n \in \mathbb{R}$. Then we have*

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \leq \sum_{k=1}^{n} a_k^2 \cdot \sum_{k=1}^{n} b_k^2.$$

Theorem 1.5.6. *Let $z, w \in \mathbb{C}$. Then we have the following properties:*

(A) $|zw| = |z| \cdot |w|$.

(B) $|z + w| \leq |z| + |w|$.

(C) $|z| - |w| \leq |z - w|$.

Proof. Let $z = x + yi$ and $w = u + vi$ for some $x, y, u, v \in \mathbb{R}$. Then we have

(A) $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$.

(B) By Corollary 1.5.5, we have $(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$.

$$\begin{aligned} |z+w|^2 &= (x+u)^2 + (y+v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| + |w|)^2. \end{aligned}$$

(C) Again, by Corollary 1.5.5, we have

$$\begin{aligned} |z-w|^2 &= (x-u)^2 + (y-v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| - |w|)^2. \end{aligned}$$

□

1.6 Euclidean Spaces

For $n \in \mathbb{N}$, we define the **Euclidean n-space** $\mathbb{R}^n$ as the set of all ordered n-tuples of real numbers, i.e,

$$\mathbb{R}^n := \{\boldsymbol{x} = (x_1, x_2, \ldots, x_n) \mid x_k \in \mathbb{R} \text{ for all } k = 1, 2, \ldots, n\}$$

where $x_1, x_2, \ldots, x_n$ are called the **coordinates** of $\boldsymbol{x}$. The elements of $\mathbb{R}^n$ with $n > 1$ are called **vectors** or *points*, which are denoted by boldface letters.

Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$. We define the following operations on $\mathbb{R}^n$:

- $\boldsymbol{x} + \boldsymbol{y} = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$.
- $\alpha \boldsymbol{x} = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n)$.

Then $\mathbb{R}^n$ is a *vector space* over $\mathbb{R}$ with the zero vector $\mathbf{0} = (0, 0, \ldots, 0)$.

Definition 1.6.1. Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. The **inner product** of $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as

$$\boldsymbol{x} \cdot \boldsymbol{y} = \sum_{k=1}^{n} x_k y_k.$$

The **norm** of $\boldsymbol{x}$ is defined as

$$\|\boldsymbol{x}\| = \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{\sum_{k=1}^{n} x_k^2}.$$

The **distance** between $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as

$$d(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\| = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}.$$

We can confirm that $d(x, y) = |x - y|$ in the case of $n = 1$.

Theorem 1.6.2. *Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$. Then we have the following properties:*

(A) $|\boldsymbol{x} \cdot \boldsymbol{y}| \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$.

(B) $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.

(C) $\|\boldsymbol{x} - \boldsymbol{y}\| \geq |\|\boldsymbol{x}\| - \|\boldsymbol{y}\||$.

Proof. **(A)** By Corollary 1.5.5, we have

$$|\boldsymbol{x} \cdot \boldsymbol{y}|^2 = \left| \sum_{k=1}^{n} x_k y_k \right|^2 \leq \sum_{k=1}^{n} x_k^2 \cdot \sum_{k=1}^{n} y_k^2 = \|\boldsymbol{x}\|^2 \cdot \|\boldsymbol{y}\|^2.$$

(B) By part **(A)**, we have

$$\begin{aligned}\|\boldsymbol{x} + \boldsymbol{y}\|^2 &= (\boldsymbol{x} + \boldsymbol{y}) \cdot (\boldsymbol{x} + \boldsymbol{y}) = \|\boldsymbol{x}\|^2 + 2\boldsymbol{x} \cdot \boldsymbol{y} + \|\boldsymbol{y}\|^2 \\ &\leq \|\boldsymbol{x}\|^2 + 2|\boldsymbol{x} \cdot \boldsymbol{y}| + \|\boldsymbol{y}\|^2 \leq \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 \\ &= (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2.\end{aligned}$$

(C) By part **(B)**, we have

$$\begin{aligned}\|\boldsymbol{x}\| = \|(\boldsymbol{x} - \boldsymbol{y}) + \boldsymbol{y}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\| + \|\boldsymbol{y}\| &\implies \|\boldsymbol{x}\| - \|\boldsymbol{y}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\|. \\ \|\boldsymbol{y}\| = \|(\boldsymbol{y} - \boldsymbol{x}) + \boldsymbol{x}\| \leq \|\boldsymbol{y} - \boldsymbol{x}\| + \|\boldsymbol{x}\| &\implies \|\boldsymbol{y}\| - \|\boldsymbol{x}\| \leq \|\boldsymbol{x} - \boldsymbol{y}\|.\end{aligned}$$

$\square$

Chapter 2

Basic Topology

2.1 Relations

A **relation** $\sim$ on a set S defines a subset R of $S \times S$, and vice versa.

$$R := \{(x, y) \in S \times S \mid x \sim y\}, \quad x \sim y \iff (x, y) \in R.$$

Definition 2.1.1. A relation $\sim$ on a set S is called an **equivalence relation** if it satisfies the following properties:

- Reflexivity: $x \sim x$ for all $x \in S$.
- Symmetry: If $x \sim y$, then $y \sim x$ for all $x, y \in S$.
- Transitivity: If $x \sim y$ and $y \sim z$, then $x \sim z$ for all $x, y, z \in S$.

Definition 2.1.2. A **partition** of a set S is a collection $\{A_\alpha\}_{\alpha \in I}$ of nonempty subsets of S such that

- $A_\alpha \cap A_\beta = \emptyset$ for all $\alpha, \beta \in I$ with $\alpha \neq \beta$.
- $\displaystyle\bigcup_{\alpha \in I} A_\alpha = S$.

Definition 2.1.3. Let $\sim$ be an equivalence relation on a set S. For each $x \in S$, the **equivalence class** of x is defined as

$$[x] := \{y \in S \mid y \sim x\}.$$

We denote the set of all equivalence classes of S by $S/\sim \, := \{[x] \mid x \in S\}$.

Theorem 2.1.4. *Let $\sim$ be an equivalence relation on a set S. For any $x, y \in S$, we have either*

$$[x] = [y] \quad \text{or} \quad [x] \cap [y] = \emptyset.$$

Therefore, $S/\sim$ is a partition of S.

Proof. Suppose $[x] \cap [y] \neq \emptyset$. Then there exists $z \in S$ such that $z \in [x]$ and $z \in [y]$. This implies that $z \sim x$ and $z \sim y$. By the symmetry and transitivity of $\sim$, we have $x \sim z$ and $z \sim y$, which implies that $x \sim y$.

Therefore, for any $w \in [x]$, we have $w \sim x$ and $x \sim y$, which implies that $w \sim y$, so $w \in [y]$. This shows that $[x] \subseteq [y]$. Similarly, we can show that $[y] \subseteq [x]$, so $[x] = [y]$. $\square$

Theorem 2.1.5. *Let $\{A_\alpha\}_{\alpha \in I}$ be a partition of a set S. Define a relation $\sim$ on S by*

$$x \sim y \iff x, y \in A_\alpha \text{ for some } \alpha \in I.$$

Then $\sim$ is an equivalence relation on S, and the equivalence classes of $\sim$ are precisely the sets A_α.

Proof. We need to show that $\sim$ is reflexive, symmetric, and transitive.

- Reflexivity: For any $x \in S$, there exists $\alpha \in I$ such that $x \in A_\alpha$, so $x \sim x$.
- Symmetry: If $x \sim y$, then there exists $\alpha \in I$ such that $x, y \in A_\alpha$, so $y \sim x$.
- Transitivity: If $x \sim y$ and $y \sim z$, then there exist $\alpha, \beta \in I$ such that $x, y \in A_\alpha$ and $y, z \in A_\beta$. Since $y \in A_\alpha \cap A_\beta$ and the sets $\{A_\alpha\}_{\alpha \in I}$ are disjoint, we must have $\alpha = \beta$, so $x, z \in A_\alpha$ and $x \sim z$.

Therefore, $\sim$ is an equivalence relation on S. The equivalence class of any $x \in S$ is precisely the set A_α containing x, so the equivalence classes of $\sim$ are exactly the sets A_α. $\square$

2.2 Functions

A *function* (or a *mapping*) is defined with 3 components.

- A **domain** X.
- A **codomain** Y.
- A **rule** f which associates each element $x \in X$ to a *unique* element $y \in Y$.

f is said to be a *function* of X into Y, which is denoted by $f : X \to Y$. For each $x \in X$, the *unique* element $y \in Y$ associated with x is denoted by $f(x)$.

Remark. Let $f : X \to Y$ be a function. For any subset $E \subseteq X$, we define the **image** of E under f as

$$f(E) \coloneqq \{f(x) \mid x \in E\} \subseteq Y.$$

Especially, if $E = X$, then $f(X) \subseteq Y$ is called the **range** of f.

Remark. Let $f : X \to Y$ be a function. For any subset $E \subseteq Y$, we define the **inverse image** of E under f as

$$f^{-1}(E) \coloneqq \{x \in X \mid f(x) \in E\} \subseteq X.$$

Definition 2.2.1. Let $f : X \to Y$ be a function. We define the properties of f as follows:

- f is **injective** (or *one-to-one*) if for any $x_1, x_2 \in X$,
$$f(x_1) = f(x_2) \implies x_1 = x_2.$$
- f is **surjective** (or *onto*) if $f(X) = Y$.
- f is **bijective** (or a *one-to-one correspondence*) if it is both injective and surjective.

Definition 2.2.2. Let $f : X \to Y$ be a bijective function. Then for any $y \in Y$, there exists a *unique* $x \in X$ such that $f(x) = y$. Therefore, we can define a function $f^{-1} : Y \to X$ by $f^{-1}(y) = x$. f^{-1} is called the **inverse function** of f.

Theorem 2.2.3. *Let $f : X \to Y$ be a bijective function. Then the inverse function $f^{-1} : Y \to X$ is bijective.*

Proof. Let $y_1, y_2 \in Y$. Suppose that $f^{-1}(y_1) = f^{-1}(y_2) = x$. Then, by Definition 2.2.2, we have both $f(x) = y_1$ and $f(x) = y_2$. Since f is a function, we have $y_1 = y_2$, which implies that f^{-1} is injective.

By the definition of inverse image, we have $f^{-1}(Y) \subseteq X$. Also, for any $x \in X$, we have $f(x) \in Y$, which implies that $x \in f^{-1}(Y)$. Therefore, we have $f^{-1}(Y) \supseteq X \implies f^{-1}(Y) = X$, which implies that f^{-1} is surjective. $\square$

Definition 2.2.4. Let $f : X \to Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ such that $f(E) \subseteq F$. Then we define the restriction functions as follows:

- The **restriction** of f to E is defined by

$$f|_E : E \to Y, \quad f|_E(x) = f(x) \quad \text{for all } x \in E.$$

- The **Corestriction** of f to F is defined by

$$f|^F : X \to F, \quad f|^F(x) = f(x) \quad \text{for all } x \in X.$$

- The **Bi-restriction** of f to E and F is defined by

$$f|_E^F : E \to F, \quad f|_E^F(x) = f(x) \quad \text{for all } x \in E.$$

We will simply call all of them as just *restriction*, if there is no ambiguity.

Theorem 2.2.5. *Let $f : X \to Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ such that $f(E) \subseteq F$. Then we have the following properties:*

(A) *If f is injective, then $f|_E^F$ is also injective.*

(B) *If $f(E) = F$, then $f|_E^F$ is surjective.*

(C) *If f is injective and $f(E) = F$, then $f|_E^F$ is bijective.*

Proof. **(C)** follows from **(A)** and **(B)**. Let $x_1, x_2 \in E$.

(A) Assume that f is injective. Then we have

$$f|_E^F(x_1) = f|_E^F(x_2) \implies f(x_1) = f(x_2) \implies x_1 = x_2.$$

(B) Assume that $f(E) = F$. Then for any $y \in F$, there exists $x \in E$ such that $f(x) = f|_E^F(x) = y$. Therefore, we have $f|_E^F(E) = F$.

□

Definition 2.2.6. Let X, Y, Z be sets and $f : X \to Y$, $g : Y \to Z$ be functions. Then we define the **composition** of f and g as the function $g \circ f : X \to Z$ defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in X.$$

Theorem 2.2.7. *Let X, Y, Z be sets and $f : X \to Y$, $g : Y \to Z$ be functions. Then we have the following properties:*

(A) *If f and g are injective, then $g \circ f$ is also injective.*

(B) *If f and g are surjective, then $g \circ f$ is also surjective.*

(C) *If f and g are bijective, then $g \circ f$ is also bijective.*

Proof. **(C)** follows from **(A)** and **(B)**. Let $x_1, x_2 \in X$.

(A) Assume that f and g are injective. Then we have

$$\begin{aligned}(g \circ f)(x_1) = (g \circ f)(x_2) &\implies g(f(x_1)) = g(f(x_2)) \\ \implies f(x_1) = f(x_2) &\implies x_1 = x_2.\end{aligned}$$

(B) Assume that f and g are surjective. Then for any $z \in Z$, there exists $y \in Y$ such that $g(y) = z$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Therefore, we have $(g \circ f)(x) = g(f(x)) = g(y) = z$.

□

Theorem 2.2.8. *Let S be a set of sets. For Sections 2.3 and 2.4, we consider the relation $\sim$ defined on S by*

$$X \sim Y \iff \textit{There exists a bijection } f : X \to Y.$$

The relation $\sim$ is an equivalence relation on S.

Proof. Let $X, Y, Z \in S$.

- **Reflexivity:** The identity function $id_X : X \to X$ defined by $id_X(x) = x$ for all $x \in X$ is bijective. Therefore, we have $X \sim X$.
- **Symmetry:** Suppose that $X \sim Y$. Then there exists a bijection $f : X \to Y$. By Theorem 2.2.3, the inverse function $f^{-1} : Y \to X$ is also bijective. Therefore, we have $Y \sim X$.
- **Transitivity:** Suppose that $X \sim Y$ and $Y \sim Z$. Then there exist bijections $f : X \to Y$ and $g : Y \to Z$. By Theorem 2.2.7, the composition $g \circ f : X \to Z$ is also bijective. Therefore, we have $X \sim Z$.

□

2.3 Finite Sets

Let $J_n := \{1, 2, \dots, n\}$ for each $n \in \mathbb{N}$. A set X is said to be **finite** if $J_n \sim X$ for some $n \in \mathbb{N}$. In this case, n is called the **cardinality** of X, denoted by $|X|$. Especially, if $X = \emptyset$, we define $|X| = 0$. X is said to be **infinite** if X is not finite.

Lemma 2.3.1. *Every subset of J_n is finite for each $n \in \mathbb{N}$.*

Proof. We will prove the lemma by mathematical induction on n. Let $P(n)$: Every subset of J_n is finite.

- **Base case:** Let $n = 1$. Then $J_1 = \{1\}$, whose only subsets are $\emptyset$ and J_1. $J_1 \sim J_1$ by $f := id_{J_1}$, and $\emptyset$ is finite by definition. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let $A \subseteq J_{n+1}$. We have two cases:
  - If $n + 1 \notin A$, then $A \subseteq J_n$, which is finite by the inductive hypothesis.
  - If $n+1 \in A$, then $A = B \cup \{n+1\}$ for some $B \subseteq J_n$. By the inductive hypothesis, B is finite, so there exists a bijection $f : J_m \to B$ for some $m \in \mathbb{N}$. Let $g : J_{m+1} \to A$ be defined by
  $$g(k) = \begin{cases} f(k) & \text{if } k \leq m, \\ n+1 & \text{if } k = m+1. \end{cases}$$
  Then g is bijective, which implies that A is finite.

  Thus $P(n + 1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. ☐

Theorem 2.3.2. *Every subset of a finite set is finite.*

Proof. Let X be a finite set and $Y \subseteq X$. Then there exists a bijection $f : J_n \to X$ for some $n \in \mathbb{N}$. Let $Z = f^{-1}(Y) \subseteq J_n$. Since $Z \subseteq J_n$, Z is finite by Lemma 2.3.1, so there exists a bijection $g : J_m \to Z$ for some $m \in \mathbb{N}$.

By Theorem 2.2.5, $f|_Z^Y$ is bijective. Then, by Theorem 2.2.7, the composition $f|_Z^Y \circ g : J_m \to Y$ is bijective. Thus Y is finite. ☐

Theorem 2.3.3. *Let X be a finite ordered set. Then X has both a least element and a greatest element.*

Proof. Since X is finite, there exists a bijection $f : J_n \to X$ for some $n \in \mathbb{N}$. The elements of X can be listed as $X = \{a_1, a_2, \ldots, a_n\}$, where $a_k = f(k)$ for each $k = 1, 2, \ldots, n$.

First, we will show that X has a least element by mathematical induction on n. Let $P(n)$: Any finite ordered set of cardinality n has a least element.

- **Base case:** Let $n = 1$. Then $X = \{a_1\}$, which has a least element a_1.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let $X = \{a_1, a_2, \dots, a_{n+1}\}$. By the inductive hypothesis, the subset $\{a_1, a_2, \dots, a_n\}$ has a least element, say a_k for some $k = 1, 2, \dots, n$. We have two cases:
  - If $a_k \le a_{n+1}$, then a_k is the least element of X.
  - If $a_{n+1} < a_k$, then a_{n+1} is the least element of X.

  Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$.

Next, we will show that X has a greatest element by mathematical induction on n. Let $Q(n)$: Any finite ordered set of cardinality n has a greatest element.

- **Base case:** Let $n = 1$. Then $X = \{a_1\}$, which has a greatest element a_1.
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. Let $X = \{a_1, a_2, \dots, a_{n+1}\}$. By the inductive hypothesis, the subset $\{a_1, a_2, \dots, a_n\}$ has a greatest element, say a_k for some $k = 1, 2, \dots, n$. We have two cases:
  - If $a_k \ge a_{n+1}$, then a_k is the greatest element of X.
  - If $a_{n+1} > a_k$, then a_{n+1} is the greatest element of X.

  Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the principle of mathematical induction, $Q(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.3.4. *Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:*

(A) *If there exists an injection $f : J_m \to J_n$, then $m \le n$.*

(B) *If there exists a surjection $f : J_m \to J_n$, then $m \ge n$.*

(C) *If there exists a bijection $f : J_m \to J_n$, then $m = n$.*

*This lemma guarantees the uniqueness of the cardinality of a finite set. If $J_m \sim X$ and $J_n \sim X$, then $J_m \sim J_n$, which implies that $m = n$ by **(C)**.*

Proof. We will prove **(A)** by mathematical induction on m, **(B)** by mathematical induction on n, and **(C)** follows from **(A)** and **(B)**.

(A) Let $n \in \mathbb{N}$ be fixed. Let $P(m)$: If there exists an injective function $f_m : J_m \to J_n$, then $m \le n$.

- **Base case:** Let $m = 1$. Then $m \le n$ holds.

- **Inductive step:** Assume that $P(m)$ is true for some $m \in \mathbb{N}$. If $n = 1$, there does not exist an injective function $f_{m+1} : J_{m+1} \to J_n$ since $f_{m+1}(1) \neq f_{m+1}(m+1)$, where $P(m+1)$ is true.

  Otherwise, suppose that such a function f_{m+1} exists. We have two cases:

  – $f_{m+1}(m+1) = n$. Let $f_m \equiv f_{m+1}|_{J_m}^{J_{n-1}}$. Then $f_m : J_m \to J_{n-1}$ is injective by Theorem 2.2.5. By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.

  – $f_{m+1}(m+1) = x \neq n$. Let $g_n : J_n \to J_n$ be defined by

$$g_n(y) = \begin{cases} x, & \text{if } y = n \\ n, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}.$$

  Then g_n is bijective. Let $f_m \equiv (g_n \circ f_{m+1})|_{J_m}^{J_{n-1}}$. Then $f_m : J_m \to J_{n-1}$ is injective by Theorems 2.2.5 and 2.2.7. By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.

  Thus $P(m+1)$ is true whenever $P(m)$ is true.

By the principle of mathematical induction, $P(m)$ is true for all $m \in \mathbb{N}$.

(B) Let $m \in \mathbb{N}$ be fixed. Let $Q(n)$: If there exists a surjective function $f_n : J_m \to J_n$, then $m \geq n$.

- **Base case:** Let $n = 1$. Then $m \geq n$ holds.
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. If $m = 1$, there does not exist a surjective function $f_{n+1} : J_m \to J_{n+1}$ since $f_{n+1}(1)$ is unique, where $Q(n+1)$ is true.

  Otherwise, suppose that such a function f_{n+1} exists. We have two cases:

  – $f_{n+1}(m) = n+1$. Let $K_{n+1} := (f_{n+1})^{-1}(\{n+1\}) \subseteq J_m$. Let $f_n : J_m \to J_n$ be defined by

$$f_n(y) = \begin{cases} f_{n+1}(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}.$$

  Then f_n is surjective. By the inductive hypothesis, we have $m-1 \geq n$, which implies that $m \geq n+1$.

  – $f_{n+1}(m) = x \neq n+1$. Let $g_{n+1} : J_{n+1} \to J_{n+1}$ be defined by

$$g_{n+1}(y) = \begin{cases} x, & \text{if } y = n+1 \\ n+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}.$$

Then g_{n+1} is bijective. Let $K_{n+1} := (g_{n+1})^{-1}(\{n+1\}) \subseteq J_m$. Let $f_n : J_m \to J_n$ be defined by

$$f_n(y) = \begin{cases} (g_{n+1} \circ f_{n+1})(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}.$$

Then f_n is surjective by Theorem 2.2.7. By the inductive hypothesis, we have $m - 1 \geq n$, which implies that $m \geq n + 1$.

Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the principle of mathematical induction, $Q(n)$ is true for all $n \in \mathbb{N}$.

□

Lemma 2.3.5. *Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:*

(A) *If $m \leq n$, then there exists an injection $f : J_m \to J_n$.*

(B) *If $m \geq n$, then there exists a surjection $f : J_m \to J_n$.*

(C) *If $m = n$, then there exists a bijection $f : J_m \to J_n$.*

This lemma is the converse of Lemma 2.3.4.

Proof. We will prove the lemma by explicitly constructing the functions.

(A) If $m \leq n$, define $f : J_m \to J_n$ by $f(i) = i$ for all $i \in J_m$. Then f is an injection.

(B) If $m \geq n$, define $f : J_m \to J_n$ by $f(i) = i$ for all $i \in J_n \subseteq J_m$ and $f(i) = n$ for all $i \in J_m \setminus J_n$. Then f is a surjection.

(C) If $m = n$, define $f : J_m \to J_n$ by $f(i) = i$ for all $i \in J_m$. Then f is a bijection.

□

Theorem 2.3.6. *Let X, Y be finite sets. Then we have the following properties:*

(A) *If there exists an injection $f : X \to Y$, then $|X| \leq |Y|$.*

(B) *If there exists a surjection $f : X \to Y$, then $|X| \geq |Y|$.*

(C) *If there exists a bijection $f : X \to Y$, then $|X| = |Y|$.*

*Moreover, the converses of **(A)**, **(B)**, and **(C)** are also true.*

*The contraposition of **(A)** is called the **pigeonhole principle**, which states that if $|X| > |Y|$, then there exists no injection $f : X \to Y$.*

Proof. Let $|X| = m$ and $|Y| = n$ for some $m, n \in \mathbb{N}$. Then there exist bijections $f_X : J_m \to X$ and $f_Y : J_n \to Y$. By Theorems 2.2.3 and 2.2.7, the composition

$$F \equiv (f_Y)^{-1} \circ f \circ f_X : J_m \to J_n$$

is injective, surjective, or bijective if f is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.4, we have the desired results:

(A) If f is injective, then F is injective, which implies that $m \leq n$.

(B) If f is surjective, then F is surjective, which implies that $m \geq n$.

(C) If f is bijective, then F is bijective, which implies that $m = n$.

Also, by Theorems 2.2.3 and 2.2.7, the composition

$$f \equiv f_Y \circ G \circ (f_X)^{-1} : X \to Y$$

is injective, surjective, or bijective if $G : J_m \to J_n$ is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.5, we have the desired converses:

(A) If $m \leq n$, then there exists an injective function $G : J_m \to J_n$, which implies that there exists an injective function $f : X \to Y$.

(B) If $m \geq n$, then there exists a surjective function $G : J_m \to J_n$, which implies that there exists a surjective function $f : X \to Y$.

(C) If $m = n$, then there exists a bijective function $G : J_m \to J_n$, which implies that there exists a bijective function $f : X \to Y$.

□

Lemma 2.3.7. *Let X, Y be finite sets with $X \cap Y = \emptyset$. Then the union $X \cup Y$ is finite.*

Proof. Let $m = |X|$ and $n = |Y|$. There exist bijections $f : J_m \to X$ and $g : J_n \to Y$. Let $h : J_{m+n} \to X \cup Y$ be defined by

$$h(i) = \begin{cases} f(i), & \text{if } i \leq m \\ g(i-m), & \text{if } m < i \leq m+n \end{cases}.$$

Then h is bijective, which implies that $X \cup Y$ is finite. □

Theorem 2.3.8. *Let X, Y be finite sets. Then the union $X \cup Y$ and intersection $X \cap Y$ are finite.*

Proof. Let A, B, C be defined by

- $A = X \setminus Y \subseteq X$.
- $B = Y \setminus X \subseteq Y$.
- $C = X \cap Y \subseteq X$.

Then A, B, C are finite by Theorem 2.3.2 and are disjoint each other. We have the followings:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$.
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$.
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$.

By Lemma 2.3.7, $X, Y, X \cup Y$ are finite. □

Theorem 2.3.9. *Let $X_1, X_2, \ldots, X_n$ be finite sets for some $n \in \mathbb{N}$ with $n \geq 2$. Then the union $\displaystyle\bigcup_{k=1}^{n} X_k$ and intersection $\displaystyle\bigcap_{k=1}^{n} X_k$ are finite.*

Proof. Since $\displaystyle\bigcap_{k=1}^{n} X_k \subseteq X_1$, $\displaystyle\bigcap_{k=1}^{n} X_k$ is finite by Theorem 2.3.2 for any $n \in \mathbb{N}$. Let $P(n) : \displaystyle\bigcup_{k=1}^{n} X_k$ is finite for some $n \in \mathbb{N}$, given $X_1, X_2, \ldots, X_n$ are finite sets.

- **Base case:** Let X_1, X_2 be finite sets. Then, by Theorem 2.3.8, $\displaystyle\bigcup_{k=1}^{2} X_k = X_1 \cup X_2$ is finite. Thus, $P(2)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

  Let $X_1, X_2, \ldots, X_{n+1}$ be finite sets. Then we have

  $$\bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^{n} X_k\right) \cup X_{n+1}.$$

  By the inductive hypothesis, $\displaystyle\bigcup_{k=1}^{n} X_k$ is finite. Then, by Theorem 2.3.8, $\displaystyle\bigcup_{k=1}^{n+1} X_k$ is finite. Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Theorem 2.3.10. *Let X be a nonempty set. Then there exists no surjection $f : X \to \mathcal{P}(X)$, where*

$$\mathcal{P}(X) = \{E \mid E \subseteq X\}$$

*is the **power set** of X. In particular, if X is finite, then $|\mathcal{P}(X)| > |X|$. This theorem is called **Cantor's theorem**.*

Proof. Let $S = \{x \in X \mid x \notin f(x)\}$. Then $S \subseteq X$, which implies that $S \in \mathcal{P}(X)$. Suppose that there exists a surjection $f : X \to \mathcal{P}(X)$. Then there exists $x \in X$ such that $f(x) = S$.

If $x \in S$, then $x \notin f(x) = S$, which is a contradiction. If $x \notin S$, then $x \in f(x) = S$, which is also a contradiction. Therefore, there exists no surjection $f : X \to \mathcal{P}(X)$.

If X is finite, by Theorem 2.3.6, there exists no surjection $f : X \to \mathcal{P}(X)$ if and only if $|X| < |\mathcal{P}(X)|$. $\square$

2.4 Countable and Uncountable Sets

A set X is said to be **countable** if $\mathbb{N} \sim X$. A set X is said to be **uncountable** if it is neither finite nor countable. A set X is said to be **at most countable** if X is either finite or countable.

Theorem 2.4.1. *Let X be an infinite set. If $X \sim Y$, then Y is infinite.*

Proof. Suppose Y is finite. Then there exists a bijection $f : J_n \to Y$ for some $n \in \mathbb{N}$. Since $X \sim Y$, there exists a bijection $g : X \to Y$.

Then the composition $g^{-1} \circ f : J_n \to X$ is a bijection, which implies that X is finite. This contradicts the assumption that X is infinite. Thus, Y is infinite. $\square$

Theorem 2.4.2. *Let X be a countable set. If $X \sim Y$, then Y is countable.*

Proof. Since X is countable, there exists a bijection $f : \mathbb{N} \to X$. Since $X \sim Y$, there exists a bijection $g : X \to Y$. Then, by Theorem 2.2.7, the composition $g \circ f : \mathbb{N} \to Y$ is a bijection, which implies that Y is countable. $\square$

Lemma 2.4.3. *$\mathbb{N}$ is infinite.*

Proof. Suppose $\mathbb{N}$ is finite. Then there exists a bijection $f : J_n \to \mathbb{N}$ for some $n \in \mathbb{N}$. Then, by Theorem 2.3.2, $f(J_n)$ is finite. By Theorem 2.3.3, $f(J_n)$ has a greatest element, say m.

Then we have $m+1 \in \mathbb{N}$ and $m < m+1 \notin f(J_n)$, which contradicts the assumption that f is surjective. Thus, $\mathbb{N}$ is infinite. □

Theorem 2.4.4. *Every countable set is infinite.*

Proof. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \to X$. By Theorem 2.4.1 and lemma 2.4.3, X is infinite. □

Remark. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \to X$. The elements of X can be listed as $X = \{x_1, x_2, x_3, \ldots\}$, where $x_n = f(n)$ for all $n \in \mathbb{N}$. In other words, the elements of X can be arranged in a sequence.

Theorem 2.4.5. *$\mathbb{Z}$ is countable.*

Proof. Let $f : \mathbb{N} \to \mathbb{Z}$ be defined by

$$f(n) = \begin{cases} -\dfrac{n}{2}, & \text{if } n \text{ is even,} \\ \dfrac{n-1}{2}, & \text{if } n \text{ is odd.} \end{cases}$$

We will show that f is bijective, which implies that $\mathbb{Z}$ is countable.

- **Injectivity:** Suppose $f(n_1) = f(n_2)$ for some $n_1, n_2 \in \mathbb{N}$. Since $f(n) \geq 0$ for odd n and $f(n) < 0$ for even n, we have n_1 and n_2 are either both even or both odd.
  - If both n_1 and n_2 are even, then $-\dfrac{n_1}{2} = -\dfrac{n_2}{2}$, which implies $n_1 = n_2$.
  - If both n_1 and n_2 are odd, then $\dfrac{n_1 - 1}{2} = \dfrac{n_2 - 1}{2}$, which implies $n_1 = n_2$.

  Thus, f is injective.
- **Surjectivity:** Let $k \in \mathbb{Z}$. We need to find an $n \in \mathbb{N}$ such that $f(n) = k$.
  - If $k \geq 0$, let $n = 2k + 1$. Then $f(n) = \dfrac{n-1}{2} = \dfrac{2k}{2} = k$.
  - If $k < 0$, let $n = -2k$. Then $f(n) = -\dfrac{n}{2} = -\dfrac{-2k}{2} = k$.

Thus, f is surjective.

Since f is both injective and surjective, we have f is bijective. $\square$

Lemma 2.4.6. *Every infinite subset of* $\mathbb{N}$ *is countable.*

Proof. Let $X \subseteq \mathbb{N}$ be an infinite subset. Let $X_1 := X$ and define a function $f : \mathbb{N} \to X$ as follows:

- $f(n) = \min X_n$
- $X_{n+1} := X_n \setminus \{f(n)\}$

for each $n \in \mathbb{N}$. By Theorem 1.1.1, X_n has a least element, so f is well-defined. We will show that f is bijective, which implies that X is countable.

- **Injectivity:** Suppose $n_1 \neq n_2$ for some $n_1, n_2 \in \mathbb{N}$. Without loss of generality, assume that $n_1 < n_2$. Then $f(n_1) \in X_{n_1}$ and $f(n_2) \in X_{n_2}$. Since $X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\}$, we have $f(n_2) \neq f(n_1)$. Thus, f is injective.
- **Surjectivity:** Let $x \in X$. Let $S := \{n \in X \mid n < x\}$. If $S = \emptyset$, then $f(1) = x$. Otherwise, by Lemma 2.3.1, $S \subseteq \{1, 2, \ldots, x-1\}$ is finite. Let $k = |S|$, $S = \{s_1, s_2, \ldots, s_k\}$ with $s_1 < s_2 < \cdots < s_k$. Then we have
  - $\min X_1 = \min S \implies X_2 = X \setminus \{s_1\}$.
  - $\min X_2 = \min S \setminus \{s_1\} \implies X_3 = X \setminus \{s_1, s_2\}$.

    $\vdots$
  - $\min X_k = \min S \setminus \{s_1, s_2, \ldots, s_{k-1}\} \implies X_{k+1} = X \setminus S$.

  Thus, $f(k+1) = \min X_{k+1} = x$. Therefore, f is surjective.

$\square$

Theorem 2.4.7. *Every infinite subset of a countable set is countable.*

Proof. Let X be a countable set and let $Y \subseteq X$ be an infinite subset. Since X is countable, there exists a bijection $f : \mathbb{N} \to X$. By Theorem 2.2.5, $f|^{Y}_{f^{-1}(Y)} : f^{-1}(Y) \to Y$ is bijective. Then, by Theorem 2.4.1, $f^{-1}(Y) \subseteq \mathbb{N}$ is infinite.

By Lemma 2.4.6, $f^{-1}(Y)$ is countable. Then there exists a bijection $g : \mathbb{N} \to f^{-1}(Y)$. The composition $f|^{Y}_{f^{-1}(Y)} \circ g : \mathbb{N} \to Y$ is bijective, which implies that Y is countable. $\square$

Theorem 2.4.8. *Let X be infinite. If there exists a surjection $f : \mathbb{N} \to X$, then X is countable.*

Proof. For each $k \in X$, let $S_k := f^{-1}(\{k\})$. Since f is surjective, $S_k \neq \emptyset$ and by Theorem 1.1.1, S_k has a least element for all $k \in X$. Let $g : X \to \mathbb{N}$ be defined by

$$g(k) = \min S_k \text{ for all } k \in X.$$

Suppose $g(k_1) = g(k_2)$ for some $k_1, k_2 \in X \implies \min S_{k_1} = \min S_{k_2}$. Since $S_i \cap S_j = \emptyset$ for all $i, j \in X$ with $i \neq j$, we have $S_{k_1} = S_{k_2} \implies k_1 = k_2$. Thus, g is injective.

Since g is injective, by Theorem 2.2.5, $g|^{g(X)} : X \to g(X)$ is bijective. Since X is infinite, $g(X) \subseteq \mathbb{N}$ is infinite by Theorem 2.4.1. Then, by Lemma 2.4.6, $g(X)$ is countable, which implies that X is countable. □

Theorem 2.4.9. *Let $\{X_n\}_{n\in\mathbb{N}}$ be a sequence of countable sets. Then $\bigcup_{n=1}^{\infty} X_n$ is countable, i.e., any countable union of countable sets is countable.*

Proof. Since each X_n is countable, there exists a bijection $f_n : \mathbb{N} \to X_n$ for each $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, we define the followings:

- $t_n = \min\left\{k \in \mathbb{N} \;\middle|\; n \leq \dfrac{k(k+1)}{2}\right\}$.
- $s_n = n - \dfrac{t_n(t_n - 1)}{2}$.

Theorem 1.2.3 guarantees that such t_n exists. Let $g : \mathbb{N} \to \bigcup_{n=1}^{\infty} X_n$ be defined by

$$g(k) = f_{s_k}(t_k - s_k + 1) \text{ for all } k \in \mathbb{N}.$$

Let $x \in \bigcup_{n=1}^{\infty} X_n$. Then there exists an $m \in \mathbb{N}$ such that $x \in X_m$. Since f_m is surjective, there exists a $l \in \mathbb{N}$ such that $f_m(l) = x$. Let

- $k = \dfrac{(l+m-1)(l+m-2)}{2} + m \in \mathbb{N}$.
- $T = \left\{z \in \mathbb{N} \;\middle|\; k \leq \dfrac{z(z+1)}{2}\right\}$.

We will show that $l + m - 2 \notin T$ and $l + m - 1 \in T$, which implies that

- $t_k = \min T = l + m - 1$.
- $s_k = k - \dfrac{t_k(t_k - 1)}{2} = m$.
- $g(k) = f_{s_k}(t_k - s_k + 1) = f_m(l + m - 1 - m + 1) = f_m(l) = x$.

Consider the following inequalities when $z = l + m - 2$ and $z = l + m - 1$:

- Let $z = l + m - 2$. Then we have
$$\begin{aligned} & k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0. \end{aligned}$$
- Let $z = l + m - 1$. Then we have
$$\begin{aligned} & k - \frac{(l+m-1)(l+m)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\ &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\ &= m - (l+m-1) = 1 - l \leq 0. \end{aligned}$$

Thus g is surjective, which implies that $\bigcup_{n=1}^{\infty} X_n$ is countable by Theorem 2.4.8. □

Corollary 2.4.10. *Let $X_1, X_2, \ldots, X_n$ be countable sets. Then $\bigcup_{i=1}^{n} X_i$ is countable, i.e., any finite union of countable sets is countable.*

Proof. Let $X_{n+1}, X_{n+2}, \ldots$ be countable sets. Then, by Theorem 2.4.9, $\bigcup_{n=1}^{\infty} X_n$ is countable. Suppose that $\bigcup_{i=1}^{n} X_i$ is finite. Then $X_1 \subseteq \bigcup_{i=1}^{n} X_i$ is finite by Theorem 2.3.2, which contradicts Theorem 2.4.4. Thus, $\bigcup_{i=1}^{n} X_i$ is infinite. Since $\bigcup_{i=1}^{n} X_i \subseteq \bigcup_{n=1}^{\infty} X_n$, by Theorem 2.4.7, $\bigcup_{i=1}^{n} X_i$ is countable. □

Theorem 2.4.11. *Suppose that A is at most countable, and, for each $a \in A$, B_a is at most countable. Then the union*

$$B = \bigcup_{a \in A} B_a$$

is at most countable, i.e., any at most countable union of at most countable sets is at most countable.

Proof. We have 4 cases to consider:

- Finite union of finite sets is finite by Theorem 2.3.9.
- Finite union of countable sets is countable by Corollary 2.4.10.
- Countable union of finite sets is countable.
- Countable union of countable sets is countable by Theorem 2.4.9.

Let us prove the 3rd case. Let A be a countable set and B_a be finite for each $a \in A$. Then, there exists a bijection $f : \mathbb{N} \to A$, and for each $a \in A$, there exists a bijection $g_a : J_{n_a} \to B_a$ for some $n_a \in \mathbb{N}$. Let $h : \mathbb{N} \to B$ be defined by

- $h(j) = g_{f(1)}(j)$ for $j = 1, 2, \cdots, n_{f(1)}$.
- $h\left(\sum_{i=1}^{k-1} n_{f(i)} + m\right) = g_{f(k)}(m)$ for each $k \in \mathbb{N}$ and $m = 1, 2, \cdots, n_{f(k)}$.

Then, h is a surjection from $\mathbb{N}$ to B. By Theorem 2.4.8, B is countable. □

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